

Lab 6

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```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
```

```
grade <- function(grades_vec) {  
  has_na <- is.na(grades_vec) # did the student miss any homework  
  if (sum(has_na)!=0) {  
    # find which homework the student missed  
    pos_na <- which(has_na==TRUE)  
    # drop just one of those homework  
    grades_vec <- grades_vec[-pos_na[1]]  
  } else {  
    # Find the lowest grades  
    min_grade <- min(grades_vec)  
    pos_min <- which(grades_vec==min_grade)  
    # Drop one of the lowest grades  
    grades_vec <- grades_vec[-pos_min[1]]  
  }  
  return(mean(grades_vec))  
}
```

```

grades_df <- read.csv("student_homework.csv")
grades_df <- grades_df %>%
  rowwise(name) %>%
  mutate(avg_grade=grade(c_across(everything())) %>%
    ungroup() %>%
    arrange(desc(avg_grade))

top_student <- grades_df[1,1]
sprintf("Q2. The top scoring student is %s", top_student)

```

```
[1] "Q2. The top scoring student is student-18"
```

```

avg_of_hw <- grades_df %>%
  summarise(across(starts_with("hw"), ~mean(replace_na(., 0))))

min_grade <- min(avg_of_hw)
lowest_hw <- which(min_grade==avg_of_hw)
sprintf("Q3. The hardest homework was hw%s", lowest_hw)

```

```
[1] "Q3. The hardest homework was hw2"
```

```

grades_df_fillna <- grades_df %>%
  mutate(across(everything(), ~replace_na(., 0)))
corr_df <- grades_df_fillna %>%
  summarise(
    across(
      starts_with("hw"),
      ~cor(grades_df_fillna$avg_grade, .)
    )
  )
max_cor <- max(corr_df)
max_cor_hw <- which(corr_df==max_cor)
sprintf("Q4. The most predictive homework to avg.score was hw%s", max_cor_hw)

```

```
[1] "Q4. The most predictive homework to avg.score was hw5"
```